MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Core, Think Harder and Challenge questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 8-9 (finishing P2)

Math Inquirers P2 Ages 8-9. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three levels. Every child starts with Core, then moves up when they are ready. Challenge questions are the ones to talk about together.
- CORE builds fluency with the main idea of the lesson. Use it for warm-ups and for children who need confidence.
- THINK HARDER goes a step further: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- CHALLENGE questions are non-routine. They ask children to find every way, work backwards, test a case, justify an answer or notice what is missing. This is the style of thinking used in gifted-education-type problems. It is not an official GEP standard, and this program is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Digit Detectives (Numbers to 1000)	6 + 8 + 9 questions
Lesson 2	Missing Digit Mysteries (Addition and Subtraction within 1000)	6 + 8 + 9 questions
Lesson 3	Legs, Wheels and Arrays (Multiplication and Division)	6 + 8 + 9 questions
Lesson 4	Pattern Machines (Patterns with Multiplication)	6 + 8 + 9 questions
Lesson 5	Mystery Number Machines (Working Backwards)	6 + 8 + 9 questions
Lesson 6	Market Day (Money)	6 + 8 + 9 questions
Lesson 7	Fair Shares (Fractions)	6 + 8 + 9 questions
Lesson 8	Shape Builders (2D and 3D Shapes)	6 + 8 + 9 questions
Lesson 9	Measure Masters (Length, Mass, Volume and Time)	6 + 8 + 9 questions
Lesson 10	Challenge Day (Mixed Puzzles)	6 + 8 + 9 questions

Lesson 1: Digit Detectives

Numbers to 1000 • Week 1

CORE

C1. What number is 5 hundreds, 3 tens and 8 ones?

C2. In 914, what is the digit 1 worth?

C3. What number is 100 more than 457?

C4. What number is 10 less than 603?

C5. Use the digit cards 3, 6 and 1 once each. Make the biggest 3-digit number.

C6. Put these numbers in order, smallest first: 409, 490, 94, 940.

THINK HARDER

T1. Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.

T2. Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?

T3. What number is 3 hundreds, 14 tens and 6 ones?

T4. Ben counts back in 100s from 980: 980, 880, 780 … He stops at the last 3-digit number. What is it? How many numbers did he say?

T5. I am a 3-digit number. My digits add up to 4. Find all the numbers I could be.

T6. Use the digit cards 2, 5 and 9 once each. Which number you can make is closest to 600?

T7. How many 3-digit numbers have 7 in the hundreds place and 0 in the ones place?

T8. On a 1000 chart with rows of 10, which number is directly below 495 (one square down)? Which number is directly above 501 (one square up)?

CHALLENGE (non-routine: justify, find every way)

X1. Use any three of the digit cards 1, 3, 5 and 7. How many different 3-digit numbers can you make?

X2. How many 3-digit numbers have digits that add up to 5?

X3. I am an odd 3-digit number. My three digits are all different and add up to 6. Find all the numbers I could be.

X4. I am between 400 and 500. My tens digit is double my ones digit. My digits add up to 10. Who am I?

X5. A book has pages numbered 1 to 120. How many digits are printed altogether?

X6. Use three of the digit cards 0, 2, 5 and 8. Which 3-digit number you can make is closest to 500?

X7. Mei uses each of the digit cards 1, 2, 3, 4, 5 and 6 once to make two 3-digit numbers. She adds them. What is the smallest total she can make?

X8. Ali counts in 5s, starting at 3: 3, 8, 13, 18 … Will he say 500? Will he say 998? Explain.

X9. Ben writes all the numbers from 300 to 399. How many times does he write the digit 3?

Lesson 2: Missing Digit Mysteries

Addition and Subtraction within 1000 • Week 1

CORE

C1. 3□5 + 214 = 579. What is the missing digit?

C2. 6□8 − 205 = 433. What is the missing digit?

C3. What number makes this true? 400 + 250 = 300 + □

C4. What must be added to 365 to make 500?

C5. What number goes in the box? □ − 150 = 600

C6. True or false? 299 + 101 = 300 + 100. Explain.

THINK HARDER

T1. Find the missing digits: 5□7 + □48 = 905

T2. Find the missing digits: 7□1 − □58 = 263

T3. What number makes this true? 538 + 199 = 540 + □

T4. A shop had 905 eggs. It sold some in the morning and 260 in the afternoon. 345 eggs were left. How many eggs were sold in the morning?

T5. What number makes this true? 900 − □ = 450 + 175

T6. 125 + 250 + □ = 1000. What is □?

T7. The numbers 20, 40, 60, 80, 100, 120, 140, 160 and 180 make a magic square. What is the magic total? Which number goes in the middle?

T8. Put 11, 12, 13, 14, 15 and 16 in a magic triangle so each side adds up to 39. Which numbers go in the corners?

CHALLENGE (non-routine: justify, find every way)

X1. Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as BIG as possible.

X2. The numbers 101 to 109 make a magic square. What is the magic total? Which number goes in the middle?

X3. I add 199 to my number, then take away 99. I get 650. What is my number?

X4. Put 100, 200, 300, 400, 500 and 600 in a magic triangle so each side adds up to 900.

X5. The same number goes in both boxes: □ + 150 = 550 − □. What is it?

X6. What is the smallest 3-digit number you can add to 386 so that the answer has three digits that are all the same?

X7. Find ALL the ways to fill the boxes: □□7 + 4□□ = 1000

X8. Box A and Box B hold 250 marbles together. Box B and Box C hold 350. Box A and Box C hold 400. How many marbles are in each box?

X9. Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as SMALL as possible.

Lesson 3: Legs, Wheels and Arrays

Multiplication and Division • Week 2

CORE

C1. How many wheels are there on 5 bicycles?

C2. There are 4 rows of chairs with 5 chairs in each row. How many chairs are there?

C3. 18 sweets are shared equally among 3 children. How many sweets does each child get?

C4. How many legs do 7 hens have?

C5. An array has 2 rows of 10 dots. Write two multiplication sentences and two division sentences for it.

C6. 25 pencils are tied in bundles of 5. How many bundles are there?

THINK HARDER

T1. There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?

T2. Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?

T3. An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?

T4. 27 children go on a boat ride. Each boat holds 4 children. How many boats are needed? How many empty seats are there?

T5. In a car park there are cars (4 wheels) and motorcycles (2 wheels). There are 8 vehicles and 22 wheels. How many cars are there?

T6. Find all the numbers from 21 to 49 that can be shared equally into 5 groups AND equally into 2 groups with nothing left over.

T7. Siti has 24 cupcakes. She puts them in boxes with the same number in every box. The boxes hold 2, 3, 4, 5 or 10 cupcakes. Which box sizes leave no cupcakes over, and how many boxes does each need?

T8. A table has 4 legs and a stool has 3 legs. Ravi counts 17 legs on the tables and stools in a room. How many tables and how many stools are there?

CHALLENGE (non-routine: justify, find every way)

X1. A shop has bicycles (2 wheels) and tricycles (3 wheels), at least one of each. There are 20 wheels altogether. Find all the ways.

X2. Crayons come in boxes of 10 and packs of 5. Mei buys 6 boxes and packs altogether, with 45 crayons. How many boxes and how many packs does she buy?

X3. A farm has ducks and cows. There are 10 heads and 28 legs. How many ducks and how many cows are there?

X4. Mei says: "I counted the legs of some hens and goats. There were 9 legs." Can she be right? Explain.

X5. A tray has muffins in rows of 5. Ben takes 3 muffins from the last row. Now there are 22 muffins on the tray. How many rows are there?

X6. I am a number between 10 and 30. I can be put into groups of 3 with none left over, and into groups of 4 with none left over. When I am put into groups of 5, 4 are left over. What number am I?

X7. A toy shop has tricycles (3 wheels) and toy cars (4 wheels), at least one of each. There are 30 wheels altogether. Find all the ways.

X8. Mr Lim has between 20 and 30 sweets. When he puts them in groups of 4, 3 are left over. When he puts them in groups of 5, 2 are left over. How many sweets does he have?

X9. A pen has hens and goats. There are 2 more hens than goats. There are 22 legs altogether. How many hens and how many goats?

Lesson 4: Pattern Machines

Patterns with Multiplication • Week 2

CORE

C1. Count in 4s. Fill in the next three numbers: 4, 8, 12, __, __, __

C2. A pattern goes red, blue, red, blue … What colour is the 10th bead?

C3. Each hand has 5 fingers. How many fingers are on 6 hands?

C4. A pattern goes star, circle, circle, star, circle, circle … What is the 9th shape?

C5. What is the rule? 3, 7, 11, 15 … Write the next two numbers.

C6. A number pattern goes 10, 20, 30 … What is the 8th number?

THINK HARDER

T1. A pattern goes A, B, B, C, A, B, B, C … What is the 25th letter?

T2. Beads go red, yellow, yellow, red, yellow, yellow … How many yellow beads are in the first 18 beads?

T3. Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?

T4. Ali counts in 4s from 4. Mei counts in 10s from 10. Which numbers up to 50 do they both say?

T5. A pattern goes 1, 2, 3, 1, 2, 3 … What is the total of the first 15 numbers?

T6. Ben's pattern is 2, 5, 8, 11 … Will 30 be in his pattern? Explain.

T7. A pattern repeats every 4 shapes. The 20th shape is a heart. What is the 40th shape? How do you know?

T8. A pattern goes square, circle, circle, square, circle, circle … Mei draws 20 shapes. How many squares does she draw?

CHALLENGE (non-routine: justify, find every way)

X1. Ali writes 1, 2, 3, 4, 1, 2, 3, 4 … What is the total of the first 20 numbers?

X2. Houses are made of sticks in a row: 1 house uses 5 sticks, 2 houses use 9, 3 houses use 13. Ben has 40 sticks. What is the most houses he can make? How many sticks are left over?

X3. A frog starts at 1 and jumps 4 each time: 1, 5, 9, 13 … Will it land on 37? Will it land on 30? Explain.

X4. A pattern repeats every 5 shapes. In each group, only the 3rd shape is a star. Which of these places have a star: 13, 23, 30, 48?

X5. Light A flashes every 3 seconds. Light B flashes every 5 seconds. They flash together at the start. In the next 30 seconds, when do they flash together?

X6. A pattern goes triangle, square, circle, circle, circle, triangle, square, circle, circle, circle … How many circles are in the first 42 shapes?

X7. Mei counts in 4s from 4. Ben counts in 5s from 5. Which is bigger: Mei's 10th number or Ben's 8th number?

X8. Siti builds towers of cubes. Tower 1 has 2 cubes, tower 2 has 5 cubes, tower 3 has 8 cubes. The towers keep growing in the same way. Which tower has 29 cubes?

X9. Frog A starts at 0 and jumps 4 each time: 4, 8, 12 … Frog B starts at 2 and jumps 3 each time: 5, 8, 11 … Which numbers up to 30 do both frogs land on?

Lesson 5: Mystery Number Machines

Working Backwards • Week 3

CORE

C1. I think of a number. I multiply it by 5. I get 45. What is my number?

C2. I think of a number. I divide it by 2. I get 8. What is my number?

C3. I think of a number. I add 35. I get 90. What is my number?

C4. I think of a number. I multiply it by 3, then add 2. I get 20. What is my number?

C5. I think of a number. I take away 10, then divide by 4. I get 5. What is my number?

C6. Mei had some stickers. She gave away 14 and has 27 left. How many did she have at first?

THINK HARDER

T1. A machine does × 4, then + 13. The OUT is 45. What went IN?

T2. Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?

T3. Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?

T4. Ali, Ben and Siti have 40 marbles altogether. Ben has 5 more marbles than Ali. Siti has 8 more marbles than Ali. How many marbles does each child have? Draw a bar for each child.

T5. A machine does × 2, then − 3. What goes IN to give OUT 17? What goes IN to give OUT 1?

T6. I think of a number. I divide it by 3, then multiply by 10. I get 50. What is my number?

T7. A bus had some people on it. At the first stop, 12 got off and 5 got on. At the second stop, 8 got off. Now there are 20 people on the bus. How many were on the bus at first?

T8. A machine multiplies by a number, then adds a number. IN 1 gives OUT 9. IN 3 gives OUT 19. What does the machine do?

CHALLENGE (non-routine: justify, find every way)

X1. I think of a number. I double it, add 6, then divide by 2. I get 10. What is my number?

X2. A machine does × 2, then − 4. Ravi puts a number in, then puts the OUT straight back in again. The final OUT is 12. What number did he put in first?

X3. Ben gives Ali 5 marbles. Now they both have 20 marbles. How many marbles did each boy have before?

X4. Mei had some money. She spent half of it on a bag. Then she spent $6 on lunch. She had $9 left. How much money did she have at first?

X5. Ali, Ben and Siti share 42 stickers. Ben gets 2 more than Ali. Siti gets twice as many as Ali. How many stickers does each child get?

X6. I think of a number. If I add 10 to it, I get the same answer as when I multiply it by 3. What is my number?

X7. Siti had some sweets. She gave Ben 1 more than half of them. Now she has 7 sweets left. How many sweets did she have at first?

X8. A machine does × 2, then − 1. Mei puts a number through the machine 3 times in a row (each OUT goes back in). The final OUT is 17. What number did she start with?

X9. Three boxes have 30 pens altogether. 5 pens are moved from box A to box B. Then 3 pens are moved from box B to box C. Now all three boxes have the same number of pens. How many pens were in each box at first?

Lesson 6: Market Day

Money • Week 3

CORE

C1. Write 275¢ in dollars and cents.

C2. Pay exactly 65¢ with the fewest coins.

C3. Pay exactly $17 with the fewest notes.

C4. A pencil costs 45¢. Mei pays with a $1 coin. How much change does she get?

C5. Ali has $20. He buys a book for $12. Does he have enough money left to buy a kite for $9?

C6. Ben had some money. He spent $6 and has $9 left. How much money did he have at first?

THINK HARDER

T1. How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?

T2. Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.

T3. A bun costs 70¢. Mei pays with a $1 coin. Which coins does she get as change, using the fewest coins?

T4. Ben has $30. He buys a book for $12 and a pen for $5. How much money does he have left? Can he also buy a game for $15?

T5. Ravi had some money. He spent $8, then he spent half of what was left. He had $6 left. How much money did Ravi have at first?

T6. Siti has 5 coins. They are all 20¢ or 50¢ coins, worth $1.60 altogether. How many of each coin does she have?

T7. Ali pays exactly 90¢ with 5 coins. Find all the sets of 5 coins he could use.

T8. A pen costs $3 more than a pencil. Together they cost $11. How much does each cost?

CHALLENGE (non-routine: justify, find every way)

X1. Ali has 10 coins. They are all 10¢ or 20¢ coins, worth $1.50 altogether. How many of each coin does he have?

X2. Mei has $1. At the snack stall: chips 45¢, sweet 15¢, jelly 25¢, biscuit 30¢, lollipop 20¢. She wants as many DIFFERENT snacks as she can. Which should she buy?

X3. Ben has 6 coins. Every coin is 10¢ or 50¢. List all the amounts he could have.

X4. Find all the ways to pay exactly $1 using only 20¢ and 50¢ coins.

X5. Ravi has $2 in 20¢ coins. Mei has $2 in 50¢ coins. Who has more coins? How many more?

X6. Three friends share the cost of a $15 present. Ali pays $2 more than Ben. Siti pays $2 more than Ali. How much does each pay?

X7. Siti had some money. She spent half of it on a book. Then she spent half of what was left on a pen. She had $5 left. How much money did she have at first?

X8. Sweets cost 5¢ each and lollipops cost 20¢ each. Ben buys 8 things altogether and spends exactly $1. How many sweets and how many lollipops does he buy?

X9. Mei has one $10 note, one $5 note, one $2 note and one $1 coin. How many different amounts can she pay exactly, using one or more of them?

Lesson 7: Fair Shares

Fractions • Week 4

CORE

C1. A cake is cut into 5 equal pieces. Ben eats 2 pieces. What fraction of the cake does he eat?

C2. How many quarters make 1 whole?

C3. Which is bigger, 1/3 or 1/6?

C4. 3/10 + 4/10 = ?

C5. Which is smaller, 5/8 or 3/8?

C6. A pizza is cut into 6 equal slices. 4 slices are eaten. What fraction of the pizza is left?

THINK HARDER

T1. Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.

T2. A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?

T3. 2/7 + ? = 1 whole

T4. Ali says: “Half of my pizza is bigger than half of your pizza.” Can he be right? Explain.

T5. Draw 3 squares. Cut each one into 4 EQUAL parts in a different way.

T6. Ravi eats 3/8 of a pizza. Ben eats 2/8 of the same pizza. How much more does Ravi eat? How much of the pizza is left?

T7. A strip is cut into 4 equal parts. Then each part is cut in half. How many parts are there now? What is each part called?

T8. Siti shades 4/12 of a shape. Then she shades 5/12 more. How much of the shape is NOT shaded?

CHALLENGE (non-routine: justify, find every way)

X1. A chocolate bar has 10 equal pieces. Ali eats 1/10 of it. Mei eats 3 more pieces than Ali. What fraction of the bar is left?

X2. I am a fraction with 1 on top. I am smaller than 1/4 but bigger than 1/7. What could I be? Find all the answers.

X3. Two fractions with bottom number 9 add up to 1 whole. One of them is 3/9 more than the other. What are they?

X4. How many different ways can you add two fractions with bottom number 6 to make 1 whole? Each fraction must be at least 1/6, and 1/6 + 5/6 and 5/6 + 1/6 count as the same way.

X5. Ali shades 1/3 of a strip of 6 equal squares. How many squares does he shade?

X6. A pizza is cut into 8 equal slices. Ben eats 1/4 of the pizza. How many slices does he eat?

X7. A cake is cut into 12 equal pieces. Mei gets 4/12. Ali and Ben share the rest, but Ali gets 2/12 more than Ben. What fraction does Ben get?

X8. A paper strip is folded in half, then in half again, then in half again. When you open it, how many equal parts are there? What fraction is each part?

X9. Siti eats 1/2 of a pizza. Then Ravi eats 1/2 of what is left. What fraction of the whole pizza is left now?

Lesson 8: Shape Builders

2D and 3D Shapes • Week 4

CORE

C1. How many quarter circles make a whole circle?

C2. How many semicircles make a whole circle?

C3. How many faces does a cube have?

C4. How many squares of ALL sizes are in a grid 2 squares long and 2 squares high? (Small ones and one big one.)

C5. A tower has 3 cubes. Ali builds 2 more towers just like it. How many cubes are there altogether?

C6. Which solid has 2 flat faces that are circles and 1 curved face?

THINK HARDER

T1. How many quarter circles make 3 semicircles?

T2. How many squares are in a grid 3 squares long and 2 squares high?

T3. A stack has 2 layers. Each layer is 3 cubes long and 2 cubes wide. How many cubes?

T4. Six squares are joined in a straight line. Can they fold into a cube? Explain.

T5. A shape is made of a square with a semicircle on each of its 4 sides, curved sides outward. How many straight sides does its outside edge have? How many curved sides?

T6. How many triangles are in a square with one line from corner to corner?

T7. Mei builds a big cube 2 cubes long, 2 wide and 2 high. She takes away 1 cube from a corner. How many cubes are left?

T8. A stack is 4 cubes long, 2 cubes wide and 2 cubes high. How many cubes?

CHALLENGE (non-routine: justify, find every way)

X1. How many squares can you find in a grid 4 squares long and 3 squares high?

X2. A square has both lines from corner to corner drawn. How many triangles can you find?

X3. Ben has 3 semicircles and 1 quarter circle, all cut from same-size circles. How many more quarter circles does he need to make exactly 2 whole circles?

X4. A big cube is built from small cubes: 3 long, 3 wide and 3 high. Its outside is painted. How many small cubes have no paint at all?

X5. Steps are made of cubes. They are 3 cubes high, then 2, then 1, and every step is 3 cubes wide from front to back. How many cubes?

X6. How many triangles are in a triangle with 3 lines from the top corner to the bottom and 1 line across the middle?

X7. A pyramid of cubes has a top layer of 1 cube, then a layer 2 by 2, then a layer 3 by 3, then a bottom layer 4 by 4. How many cubes are there?

X8. How many squares can you find in a grid 5 squares long and 5 squares high?

X9. On a dice, opposite faces add up to 7. You can see 3 faces showing 1, 2 and 3. What numbers are on the 3 faces you cannot see? What do they add up to?

Lesson 9: Measure Masters

Length, Mass, Volume and Time • Week 5

CORE

C1. One bag of flour balances 2 bags of sugar. A bag of sugar is 300 g. How heavy is the bag of flour?

C2. A bucket holds 10 ℓ. It has 4 ℓ of water in it. Ali pours in 3 ℓ more. How many more litres are needed to fill it?

C3. A ribbon is 90 cm long. Mei cuts off 25 cm. How long is the piece that is left?

C4. A cartoon starts at 5:40 p.m. and lasts 30 minutes. What time does it end?

C5. Write a.m. or p.m. Ben eats lunch at 12:30 ____. He goes to bed at 8:45 ____.

C6. The minute hand points to 7. How many minutes past the hour is it?

THINK HARDER

T1. 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?

T2. You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.

T3. Mei's piano lesson ends at 4:15 p.m. It lasted 45 minutes. What time did it start?

T4. Buses to the MRT station leave every 15 minutes. The first bus leaves at 6:50 a.m. Ravi reaches the bus stop at 7:40 a.m. How long does he wait? The ride takes 25 minutes. When does he reach the MRT station?

T5. Three ribbons, each 30 cm long, are glued end to end in a line. Each join overlaps by 2 cm. How long is the long ribbon?

T6. A tank has 20 ℓ of water. A tap adds 5 ℓ every 10 minutes. The tap is turned on at 9:40 a.m. At what time does the tank have 40 ℓ?

T7. A box with 4 same books in it is 1000 g. The empty box is 200 g. How heavy is one book?

T8. Ali's clock is 10 minutes fast. It shows 3:05 p.m. What is the real time?

CHALLENGE (non-routine: justify, find every way)

X1. 1 melon balances 2 pineapples. 1 pineapple balances 3 apples. 1 apple balances 2 lemons. How many lemons balance 1 melon?

X2. A rope 40 cm long is cut into pieces 10 cm long. Each cut takes 2 minutes. How long does the cutting take?

X3. A film is 1 hour 40 minutes long. It ends at 12:20 p.m. What time did it start?

X4. Ali's watch loses 5 minutes every hour. He sets it to the right time at 8:00 a.m. What does his watch show at 12 noon?

X5. A basket with 4 apples is 700 g. The same basket with 2 apples is 500 g. How heavy is the empty basket?

X6. Ravi has a 5 ℓ jug, a 3 ℓ jug and a big bucket. How can he get exactly 7 ℓ of water in the bucket?

X7. You have a 3 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 5 ℓ? List each pour.

X8. Scale 1: 2 cubes and 1 ball balance 2 balls. Scale 2: 1 ball and 1 cube balance a 300 g weight. How heavy is one cube? How heavy is one ball?

X9. Mei has 3 lessons in a row. Each lesson is 35 minutes long, with a 10-minute break between lessons. The first lesson starts at 8:15 a.m. What time does the last lesson end?

Lesson 10: Challenge Day

Mixed Puzzles • Week 5

CORE

C1. Use the digit cards 5, 1 and 9 once each. Make the greatest 3-digit number and the smallest 3-digit number.

C2. In a picture graph, each ★ stands for 2 children. Apples has 4 stars. How many children chose apples?

C3. Ali has 3 bags with 5 marbles in each bag. Mei has 4 bags with 4 marbles in each bag. Who has more marbles? How many more?

C4. Make 80¢ with exactly 3 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins.

C5. A cake is cut into 8 equal pieces. Siti eats 2 pieces and Ben eats 3 pieces. What fraction of the cake is left?

C6. Ben is taller than Ali. Ali is taller than Mei. Siti is taller than Ben. Who is the tallest? Who is the shortest?

THINK HARDER

T1. 30 children chose a fruit. In the picture graph each ● stands for 5 children. Apple has ●● and banana has ●●●. The mango row is missing. How many ● should the mango row have?

T2. How many 3-digit numbers have digits that add up to 3? List them all.

T3. Ravi has 4 boxes of 5 pencils and 3 boxes of 10 pencils. He shares all the pencils equally among 5 friends. How many does each friend get?

T4. 5, 9, 13, 17, … What is the 10th number in this pattern?

T5. A farm has chickens and cows. There are 8 heads and 22 legs. How many cows are there?

T6. Mei spent half of her money on a book. Then she spent $4 on a pen. She has $6 left. How much money did she have at first?

T7. Ali, Ben and Siti each chose a different colour: red, blue or green. Ali did not choose red. Siti chose blue. Who chose green?

T8. A rope 30 m long is cut into pieces 5 m long. How many pieces are there? How many cuts?

CHALLENGE (non-routine: justify, find every way)

X1. Two numbers add up to 100. One number is 30 more than the other. What are the two numbers?

X2. I am a 3-digit number. My ones digit is 0. My hundreds digit is 2 more than my tens digit. My digits add up to 12. What number am I?

X3. There are 8 bicycles and tricycles in a park. They have 20 wheels altogether. How many tricycles are there?

X4. Ali has 9 coins. They are all 10¢ and 20¢ coins. Together they are worth $1.30. How many 20¢ coins does he have?

X5. Ali, Ben and Mei are 7, 8 and 9 years old. Ali is not the oldest. Mei is older than Ali. Ben is younger than Ali. How old is each child?

X6. A ribbon is cut into 10 equal pieces. Siti uses 3/10 of it and Ben uses 4/10 of it. What fraction is left? Each piece is 5 cm long. How long is the part that is left?

X7. A picture graph has lost its key. Football has 6 pictures and swimming has 4 pictures. 10 more children chose football than swimming. How many children chose swimming? Football?

X8. Ali, Ben, Mei and Siti each have one coin: 10¢, 20¢, 50¢ or $1. Mei has more money than Ali but less than Ben. Ben does not have the $1 coin. Who has which coin?

X9. At a hawker centre, chicken rice costs $4 a plate and noodles cost $3 a bowl. Ravi's family buys 7 dishes for $25. How many plates of chicken rice do they buy?

Answer key (tutor only)

Lesson 1: Digit Detectives

C1. 538
C2. 10 (1 ten)
C3. 557
C4. 593
C5. 631
C6. 94, 409, 490, 940
T1. 6 numbers: 158, 185, 518, 581, 815, 851.
T2. 3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.
T3. 446 (14 tens = 1 hundred and 4 tens).
T4. 180. He said 9 numbers: 980, 880, 780, 680, 580, 480, 380, 280, 180.
T5. 10 numbers: 103, 112, 121, 130, 202, 211, 220, 301, 310, 400.
T6. 592 (only 8 away).
T7. 10: 700, 710, 720, … 790.
T8. 505 is directly below 495. 491 is directly above 501.
X1. 24. Leave out one card (4 choices); each set of 3 cards makes 6 numbers: 6 + 6 + 6 + 6 = 24.
X2. 15. Hundreds digit 1: 5 numbers; 2: 4; 3: 3; 4: 2; 5: 1. 5 + 4 + 3 + 2 + 1 = 15.
X3. 6 numbers: 105, 123, 213, 231, 321, 501.
X4. 442 (4 + 4 + 2 = 10, and 4 is double 2).
X5. 252. Pages 1 to 9: 9 digits. Pages 10 to 99: 90 pages × 2 = 180 digits. Pages 100 to 120: 21 pages × 3 = 63 digits. 9 + 180 + 63 = 252.
X6. 502 (only 2 away).
X7. 381, for example 135 + 246. The smallest digits 1 and 2 go in the hundreds places, 3 and 4 in the tens, 5 and 6 in the ones: 300 + 70 + 11 = 381.
X8. He says only numbers ending in 3 or 8. So he will not say 500, but he will say 998.
X9. 120. Hundreds place: 100 times. Tens place (330 to 339): 10 times. Ones place (303, 313, … 393): 10 times. 100 + 10 + 10 = 120.

Lesson 2: Missing Digit Mysteries

C1. 6 (365 + 214 = 579)
C2. 3 (638 − 205 = 433)
C3. 350
C4. 135
C5. 750
C6. True. 299 is 1 less than 300 and 101 is 1 more than 100, so both sides are 400.
T1. 557 + 348 = 905 (ones 7 + 8 = 15; tens 5 + 4 + 1 = 10; hundreds 5 + 3 + 1 = 9).
T2. 721 − 458 = 263. Check: 263 + 458 = 721.
T3. 197. 538 is 2 less than 540, so □ is 2 less than 199. Both sides are 737.
T4. 300. Work backwards: 345 + 260 = 605 eggs after the morning sales; 905 − 605 = 300.
T5. 275 (450 + 175 = 625 and 900 − 625 = 275).
T6. 625
T7. 300 (all nine add up to 900, shared equally by the 3 rows). The middle is 100.
T8. 11, 12 and 13. Between 11 and 12: 16. Between 12 and 13: 14. Between 13 and 11: 15.
X1. 531: 654 − 123.
X2. 315 (all nine add up to 945, shared by 3 rows). The middle is 105.
X3. 550. Work backwards: 650 + 99 = 749, then 749 − 199 = 550.
X4. Corners 100, 200, 300. Between 100 and 200: 600. Between 200 and 300: 400. Between 300 and 100: 500.
X5. 200 (200 + 150 = 350 and 550 − 200 = 350).

X6. 169 (386 + 169 = 555). 444 is only 58 more than 386, and 58 is not a 3-digit number.

X7. 10 ways: 507 + 493, 517 + 483, 527 + 473, 537 + 463, 547 + 453, 557 + 443, 567 + 433, 577 + 423, 587 + 413, 597 + 403. Ones: 7 + 3 = 10. Hundreds: 5 + 4 + 1 = 10. The two tens digits add up to 9 (with the carried 1 they make 10).

X8. A 150, B 100, C 250. B + C is 100 more than A + B, so C has 100 more than A. A + C = 400, so A = 150 and C = 250. Then B = 250 − 150 = 100.

X9. 47: 412 − 365. The hundreds digits are next to each other (4 and 3); then make 4□□ as small as possible (412) and 3□□ as big as possible (365).

Lesson 3: Legs, Wheels and Arrays

C1. 10 wheels (2, 4, 6, 8, 10).

C2. 20 chairs (4 × 5 = 20).

C3. 6 sweets (18 ÷ 3 = 6).

C4. 14 legs (7 × 2 = 14).

C5. 2 × 10 = 20, 10 × 2 = 20, 20 ÷ 2 = 10, 20 ÷ 10 = 2.

C6. 5 bundles (25 ÷ 5 = 5).

T1. 6 hens and 3 goats. Start with 9 hens (18 legs); each swap to a goat adds 2 legs; 3 swaps make 24. Check: 12 + 12 = 24.

T2. 8 each (4 × 8 = 32) and 2 left over.

T3. 16 stickers. 3 rows have 12, so each row has 4 (12 ÷ 3 = 4); 4 rows of 4 is 16.

T4. 7 boats (6 boats hold 24, and the other 3 children need a boat too). 7 boats have 28 seats, so 1 empty seat.

T5. 3 cars (and 5 motorcycles). Start with 8 motorcycles (16 wheels); each swap to a car adds 2 wheels; 3 swaps make 22. Check: 12 + 10 = 22.

T6. 30 and 40 (they are in both the 5 and the 2 times tables).

T7. Boxes of 2 (12 boxes), boxes of 3 (8 boxes), boxes of 4 (6 boxes). Boxes of 5 and 10 leave 4 over.

T8. 2 tables and 3 stools (8 + 9 = 17). No other mix works: 0, 1, 3 or 4 tables leave 17, 13, 5 or 1 legs, which are not in the 3 times table.

X1. 3 ways: 7 bicycles and 2 tricycles; 4 bicycles and 4 tricycles; 1 bicycle and 6 tricycles.

X2. 3 boxes and 3 packs (30 + 15 = 45).

X3. 6 ducks and 4 cows (12 + 16 = 28 legs).

X4. No. Hens have 2 legs and goats have 4, so the total is always an even number (in the 2 times table). 9 is odd.

X5. 5 rows. Before Ben took 3 there were 25 muffins, and 25 is 5 rows of 5.

X6. 24. Numbers in both the 3 and 4 times tables between 10 and 30 are 12 and 24; 12 in groups of 5 leaves 2, 24 leaves 4.

X7. 2 ways: 6 tricycles and 3 cars (18 + 12); 2 tricycles and 6 cars (6 + 24). With 1, 2, 4, 5 or 7 cars the wheels left are not in the 3 times table.

X8. 27. Groups of 4 with 3 left: 23 or 27. Groups of 5 with 2 left: 22 or 27. Only 27 is in both lists.

X9. 5 hens and 3 goats. Guess and check: 1 goat and 3 hens make 10 legs; 2 and 4 make 16; 3 and 5 make 22. Check: 12 + 10 = 22.

Lesson 4: Pattern Machines

C1. 16, 20, 24

C2. Blue (every even place is blue).

C3. 30 fingers (5, 10, 15, 20, 25, 30).

C4. A circle. 9 ends a group of 3, and each group ends with a circle.

C5. Add 4. The next numbers are 19 and 23.

C6. 80

T1. A. 24 ends a group of 4 (4 × 6 = 24), so 25 starts again with A.

T2. 12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).

T3. 25 sticks. The rule is add 4: 5, 9, 13, 17, 21, 25.

T4. 20 and 40.

T5. 30. 15 numbers is 5 groups, and each group adds up to 6: 6 + 6 + 6 + 6 + 6 = 30.

T6. No. The pattern goes 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32 … It jumps from 29 to 32. (Every number is 1 less than a number in the 3 times table, and 30 is in the 3 times table.)

T7. A heart. 20 and 40 are both in the 4 times table, so both are the last shape of a group.

T8. 7 squares (in places 1, 4, 7, 10, 13, 16, 19). 18 shapes is 6 groups with 6 squares, and place 19 is one more square.

X1. 50. 20 numbers is 5 groups, and each group adds up to 10.

X2. 9 houses use 37 sticks, with 3 left over (10 houses would need 41).

X3. It lands on 37 but not on 30. The frog lands on numbers 1 more than the 4 times table: 36 + 1 = 37, but 30 is not 1 more than a 4 times table number (it lands on 29 and 33).

X4. 13, 23 and 48. Stars are 2 more than the 5 times table: 13 = 10 + 3, 23 = 20 + 3, 48 = 45 + 3. 30 ends a group, so it is the 5th shape.

X5. At 15 seconds and at 30 seconds (numbers in both the 3 and 5 times tables).

X6. 24 circles. 40 shapes is 8 groups with 3 circles each (24); places 41 and 42 are a triangle and a square.

X7. Neither: they are the same. Mei's 10th number is 40 and Ben's 8th number is 40.

X8. Tower 10. The rule is add 3: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29.

X9. 8 and 20. Frog A: 4, 8, 12, 16, 20, 24, 28. Frog B: 5, 8, 11, 14, 17, 20, 23, 26, 29.

Lesson 5: Mystery Number Machines

C1. 9

C2. 16

C3. 55

C4. 6 (20 − 2 = 18, 18 ÷ 3 = 6)

C5. 30 (5 × 4 = 20, 20 + 10 = 30)

C6. 41 (27 + 14)

T1. 8 (45 − 13 = 32, 32 ÷ 4 = 8)

T2. $47 (42 − 10 = 32, 32 + 15 = 47)

T3. 36 (30 − 12 = 18 is half, 18 + 18 = 36)

T4. Ali 9, Ben 14, Siti 17 (40 − 5 − 8 = 27, 27 ÷ 3 = 9; check 9 + 14 + 17 = 40)

T5. 10, and 2 (17 + 3 = 20, 20 ÷ 2 = 10; 1 + 3 = 4, 4 ÷ 2 = 2)

T6. 15 (50 ÷ 10 = 5, 5 × 3 = 15)

T7. 35 (20 + 8 = 28, 28 − 5 = 23, 23 + 12 = 35)

T8. × 5, then + 4 (1 × 5 + 4 = 9 and 3 × 5 + 4 = 19; × 2, × 3, × 4 and × 10 do not fit both)

X1. 7 (10 × 2 = 20, 20 − 6 = 14, 14 ÷ 2 = 7)

X2. 6 (12 + 4 = 16, 16 ÷ 2 = 8; 8 + 4 = 12, 12 ÷ 2 = 6; check 6 → 8 → 12)

X3. Ben 25, Ali 15

X4. $30 (9 + 6 = 15 is half, 15 + 15 = 30)

X5. Ali 10, Ben 12, Siti 20 (Ali's share 4 times, plus 2, makes 42: 40 ÷ 4 = 10)

X6. 5 (5 + 10 = 15 and 5 × 3 = 15)

X7. 16 (7 + 1 = 8 is half, 8 + 8 = 16; she gave Ben 9)

X8. 3 (17 + 1 = 18, 18 ÷ 2 = 9; 9 + 1 = 10, 10 ÷ 2 = 5; 5 + 1 = 6, 6 ÷ 2 = 3; check 3 → 5 → 9 → 17)

X9. A 15, B 8, C 7 (each ends with 10; undo the moves from the end)

Lesson 6: Market Day

C1. $2.75

C2. 50¢ + 10¢ + 5¢ (3 coins)

C3. $10 + $5 + $2 (3 notes)

C4. 55¢

C5. No. He has $8 left, which is $1 short.

C6. $15

T1. 10 ways (two $5: 1 way; one $5: 3 ways; no $5: 6 ways)

T2. 2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢

T3. 30¢: 20¢ + 10¢

T4. $13 left. No, he is $2 short.

T5. $20 ($6 + $6 = $12 was left after the $8; $12 + $8 = $20)
T6. Three 20¢ and two 50¢ (60¢ + 100¢ = 160¢)
T7. 3 sets: 50¢ + 20¢ + 10¢ + 5¢ + 5¢; 50¢ + 10¢ + 10¢ + 10¢ + 10¢; 20¢ + 20¢ + 20¢ + 20¢ + 10¢
T8. Pencil $4, pen $7 ($11 − $3 = $8, $8 ÷ 2 = $4)
X1. Five 10¢ and five 20¢ (start with ten 10¢ = $1; each swap to a 20¢ adds 10¢; 5 swaps)
X2. Sweet, lollipop, jelly and biscuit (15 + 20 + 25 + 30 = 90¢, 10¢ change). Five snacks cost $1.35, too much.
X3. 7 amounts: 60¢, $1, $1.40, $1.80, $2.20, $2.60, $3
X4. 2 ways: two 50¢, or five 20¢ (one 50¢ leaves 50¢, which 20¢ coins cannot make)
X5. Ravi, 6 more (10 coins and 4 coins)
X6. Ben $3, Ali $5, Siti $7
X7. $20 ($5 + $5 = $10 before the pen; $10 + $10 = $20)
X8. 4 sweets and 4 lollipops (20¢ + 80¢ = $1)
X9. 15 amounts: $1, $2, $3, $5, $6, $7, $8, $10, $11, $12, $13, $15, $16, $17, $18

Lesson 7: Fair Shares

C1. 2/5
C2. 4
C3. 1/3 (the same whole cut into 3 parts gives bigger parts than cut into 6)
C4. 7/10
C5. 3/8
C6. 2/6
T1. 1/12, 1/5, 1/3, 1/2
T2. 2/9 (4 + 3 = 7 pieces used, 9 − 7 = 2 left)
T3. 5/7
T4. Yes, if his pizza is bigger. Halves of different-size wholes are different sizes.
T5. Any 3 of: 4 strips across, 4 strips down, a 2 by 2 grid, both lines from corner to corner (4 same triangles). Every part must be the same size.
T6. 1/8 more; 3/8 is left (5 eighths eaten).
T7. 8 parts; each part is one eighth (1/8). 2 eighths make 1 quarter.
T8. 3/12 (9 twelfths shaded, 12 − 9 = 3)
X1. 5/10 (Ali 1 piece, Mei 4 pieces: 5 eaten, 5 left)
X2. 1/5 and 1/6
X3. 6/9 and 3/9 (take away the extra 3 ninths: 6 left; share: 3 and 3; give the 3 back: 6 and 3)
X4. 3 ways: 1/6 + 5/6, 2/6 + 4/6, 3/6 + 3/6
X5. 2 squares (split the 6 squares into 3 equal groups of 2; one group is 1/3)
X6. 2 slices (2 eighths make 1 quarter; show it with bars. Equivalent fractions are a P3 idea, so this is a stretch)
X7. 3/12 (and Ali gets 5/12): 8 pieces are shared; take away Ali's extra 2, share 6 as 3 and 3.
X8. 8 equal parts; each is 1/8 (1 fold: 2 parts, 2 folds: 4, 3 folds: 8).
X9. 1/4. After Siti, half is left; Ravi eats half of that half, which is a quarter of the pizza, so a quarter is left.

Lesson 8: Shape Builders

C1. 4
C2. 2
C3. 6
C4. 5 (4 small + 1 big)
C5. 9
C6. A cylinder
T1. 6
T2. 8 (6 small + 2 made of 4 small ones)
T3. 12
T4. No. When you fold the line round, the 5th square lands on the same face as the 1st and the 6th on the 2nd, and the two ends are left open.

T5. No straight sides and 4 curved sides.
T6. 2
T7. 7
T8. 16 (8 in each layer)
X1. 20 (12 + 6 + 2)
X2. 8 (4 small + 4 made of 2 small ones)
X3. 1 more (3 semicircles = 6 quarter circles, + 1 = 7; 2 circles need 8)
X4. 1 (only the cube in the very middle is not on the outside)
X5. 18 (9 + 6 + 3)
X6. 20 (10 sit on the bottom line + 10 sit on the line across)
X7. 30 (1 + 4 + 9 + 16)
X8. 55 (25 + 16 + 9 + 4 + 1)
X9. 6, 5 and 4; they add up to 15.

Lesson 9: Measure Masters

C1. 600 g (300 + 300)
C2. 3 ℓ (4 + 3 = 7, and 10 − 7 = 3)
C3. 65 cm
C4. 6:10 p.m. (5:40 to 6:00 is 20 minutes, 10 more minutes)
C5. p.m. and p.m.
C6. 35 minutes (count in 5s: 5, 10, 15, 20, 25, 30, 35)
T1. Swap the pomelo for 3 apples: 5 apples balance 500 g, so one apple is 100 g. The pomelo is 300 g. Check: 300 + 100 + 100 = 500.
T2. Fill the 7 ℓ jug (0, 7). Pour into the 2 ℓ jug (2, 5). Empty the 2 ℓ jug (0, 5). Pour again (2, 3). 3 ℓ is left in the 7 ℓ jug.
T3. 3:30 p.m. (count back 15 minutes to 4:00, then 30 more minutes to 3:30)
T4. Buses: 6:50, 7:05, 7:20, 7:35, 7:50. He waits 10 minutes for the 7:50 bus and reaches the station at 8:15 a.m. (7:50 to 8:00 is 10 minutes, 15 more minutes).
T5. 86 cm (30 + 30 + 30 = 90; there are 2 joins, so take away 2 + 2 = 4)
T6. 10:20 a.m. (20 ℓ more is 4 lots of 5 ℓ, which takes 4 lots of 10 minutes = 40 minutes)
T7. 200 g (the books are 1000 − 200 = 800 g; 200 + 200 + 200 + 200 = 800)
T8. 2:55 p.m.
X1. 12 lemons (1 melon = 6 apples, and each apple = 2 lemons)
X2. 6 minutes (4 pieces need only 3 cuts)
X3. 10:40 a.m. (1 hour before 12:20 p.m. is 11:20 a.m., then 40 minutes earlier is 10:40 a.m.)
X4. 11:40 a.m. (4 hours, 5 minutes slow each hour: 20 minutes slow)
X5. 300 g (the 2 extra apples are 200 g, so each apple is 100 g; 500 − 200 = 300)
X6. One way: fill the 5 ℓ jug and pour it into the 3 ℓ jug until that is full: 2 ℓ is left. Pour the 2 ℓ into the bucket. Then fill the 5 ℓ jug and pour it all into the bucket: 2 + 5 = 7 ℓ. (Other ways work too, e.g. 1 ℓ + 3 ℓ + 3 ℓ. Whole jugs alone give 3, 5, 6, 8, 9, 10 ..., never 7.)
X7. (3 ℓ, 7 ℓ): fill 7 (0, 7); pour (3, 4); empty 3 (0, 4); pour (3, 1); empty 3 (0, 1); pour (1, 0); fill 7 (1, 7); pour until the 3 ℓ jug is full (3, 5). 5 ℓ in the 7 ℓ jug.
X8. Take one ball off both sides of scale 1: 2 cubes balance 1 ball. On scale 2, swap the ball for 2 cubes: 3 cubes balance 300 g, so a cube is 100 g and a ball is 200 g. Check scale 1: 100 + 100 + 200 = 400 = 200 + 200.
X9. 10:20 a.m. (8:15 to 8:50, break to 9:00, 9:00 to 9:35, break to 9:45, 9:45 to 10:20)

Lesson 10: Challenge Day

C1. 951 and 159
C2. 8 children
C3. Mei, by 1 (Ali 15, Mei 16)
C4. 50¢ + 20¢ + 10¢ (the only way)
C5. 3/8
C6. Siti is the tallest. Mei is the shortest.

T1. 1 ● (apple 10, banana 15, so mango is 30 − 25 = 5 children)

T2. 6: 102, 111, 120, 201, 210, 300

T3. 10 (20 + 30 = 50 pencils; 50 ÷ 5 = 10)

T4. 41 (add 4: 21, 25, 29, 33, 37, 41)

T5. 3 cows and 5 chickens (12 + 10 = 22)

T6. $20 (the half she kept is $4 + $6 = $10)

T7. Ali (Siti has blue, Ali is not red, so Ali is green and Ben is red)

T8. 6 pieces and 5 cuts

X1. 35 and 65 (take away the extra 30: 70, share: 35 and 35, give 30 back)

X2. 750

X3. 4 tricycles and 4 bicycles (12 + 8 = 20)

X4. 4 (four 20¢ = 80¢ and five 10¢ = 50¢)

X5. Ben 7, Ali 8, Mei 9

X6. 3/10 is left; 15 cm

X7. Swimming 20, football 30 (2 pictures stand for 10 children, so 1 picture stands for 5)

X8. Ali 10¢, Mei 20¢, Ben 50¢, Siti $1. Ben is not $1 and two people have less than him, so Ben has 50¢; then Ali 10¢ and Mei 20¢, and Siti has the $1.

X9. 4 plates of chicken rice and 3 bowls of noodles ($16 + $9 = $25). Table: 0 → $21, 1 → $22, 2 → $23, 3 → $24, 4 → $25 ✓